

AERN Integration Guide

Deployment, Initialization, Command Reference, and Tuning Manual

Authenticated Encrypted Relay Network

Revision 1.1 - May 22, 2026

Document status: This operations guide defines the deployment, initialization, command, and tuning procedures for an AERN operating domain. Where executable names differ between builds, substitute the local role binary or service wrapper used by the deployment package.

Field	Value
System	AERN - Authenticated Encrypted Relay Network
Document type	Operations and deployment manual
Primary audience	Network administrators, cryptographic operators, deployment engineers, and validation teams
Authoring basis	AERN source headers, implementation behavior, task validation results, AERN specification, and formal analysis
Security model	Controlled-domain, certificate-anchored, post-quantum-ready relay network

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1. Overview and operational model

AERN is an authenticated encrypted relay network intended for controlled proxy domains. It is not a public volunteer anonymity network. Its operational model is based on a root trust anchor, a domain controller, authenticated proxy servers, client enrollment, and pre-established encrypted tunnels between authorized relay peers.

The deployed network has four administrative phases. First, the ARS creates and protects the root signing key and root certificate. Second, the ADC is certified by the ARS and becomes the domain control authority for topology, registration, revocation, convergence, and remote signing. Third, APS proxy servers are certified, registered, synchronized, and mutually keyed. Fourth, clients register or join the domain, select an entry APS, and establish an encrypted client-to-entry tunnel.

The data phase uses fixed-size relay packets and encrypted route and session metadata. Per-hop APS tunnels authenticate the visible outer packet header as associated data. The route path, relay payload header, session identifiers, fragment fields, payload type, return flag, and backend payload remain inside the encrypted relay plaintext.

1.1 Operational objectives

Objective	Operational requirement
Authenticated membership	Every ARS, ADC, APS, and client identity must be represented by a certificate that chains to the accepted root.
Controlled topology	The ADC maintains and distributes the topology. Proxy route hints are meaningful only relative to synchronized topology order.
Efficient relay transport	Asymmetric operations are constrained to registration, certificates, and tunnel establishment. Regular relay packets use symmetric authenticated encryption.
Metadata protection	Relay metadata is encrypted inside the tunnel plaintext. Operational logs must not record client content, route paths, session identifiers, packet identifiers, fragment identifiers, or backend payload bytes.
Low-latency anonymity	AERN uses fixed-size packets, per-packet interior route changes, dummy packets, and randomized ingress delay, but it does not claim mixnet-grade protection against a fully global timing adversary.

2. Roles, trust boundaries, and deployment layout

Role	Primary responsibility	Operational notes
ARS	Root certificate authority and trust anchor.	Should be isolated. Network service should be restricted to remote signing when enabled.
ADC	Domain controller for registration, topology, revocation, convergence, and administrative messages.	Must be certified by ARS. Controls the operational topology and distributes signed updates.
APS	Authenticated proxy server in the relay mesh.	Maintains peer tunnel states, relay sessions, fragment cache, pending/delay queues, dummy traffic counters, and backend callbacks.
Client	Authenticated endpoint that enters the AERN domain through an APS.	Obtains APS topology from ADC and establishes a client-to-entry tunnel.
Backend adapter	Local transport integration point at ingress or egress.	Outside the cryptographic relay proof. Must not log or export relay metadata.

2.1 Recommended reference topology

A minimum validation topology should contain one ARS, one ADC, four APS nodes, one client, and one simulated or real backend destination. A production topology should normally use at least three route-addressable APS nodes because the route generator requires an ingress, at least one forwarding node, and an egress.

Reference domain:

```
ARS  root-aern.example.net  10.10.0.10:38764
ADC  adc-aern.example.net   10.10.0.20:38762
APS1 aps-01.example.net    10.10.1.11:38766
```

```

APS2 aps-02.example.net    10.10.1.12:38766
APS3 aps-03.example.net    10.10.1.13:38766
APS4 aps-04.example.net    10.10.1.14:38766
ACD  client-01             dynamic:37761
DST  https-service         203.0.113.10:443

```

2.2 Trust boundary summary

Boundary	Protected by	Failure consequence
ARS to certificate	Root signature over certificate hash.	Invalid certificate must not be accepted.
ADC to topology/control message	ADC certificate and message signature over payload, sequence, timestamp, and version where implemented.	Rejected administrative message; topology must not be updated on failure.
APS tunnel	KEM-derived directional RCS state and authenticated outer header.	Packet rejection, auth-failure accounting, and possible tunnel teardown at threshold.
Relay session	Encrypted session-open/session-ack payloads and session cache state.	No backend delivery before egress acknowledgement.
Fragment cache	Authenticated relay fragment metadata and timeout policy.	No delivery until complete and consistent fragment set.
Backend	Local callback contract.	AERN cryptographic proof ends at serialized packet handoff.

3. Build and installation prerequisites

Build the AERN application and tests together with the QSC library configured for the selected cryptographic parameter set. AERN depends on QSC primitives, including SHA-3/SHAKE, KMAC, RCS, and the configured post-quantum KEM and signature families.

3.1 Source package layout

```

workspace/
  AERN/      # AERN library and application role modules
  AERNTTest/ # test harness and in-process validation code
  QSC/       # cryptographic and utility library dependency

```

3.2 Visual Studio build sequence

1. Open or create a Visual Studio solution containing:
 - QSC.vcxproj
 - AERN.vcxproj
 - AERNTTest.vcxproj
2. Build QSC first.
3. Build AERN next, with the AERN include path and QSC include path configured.
4. Build AERNTTest last, with references to both AERN and QSC.
5. Run AERNTTest before attempting a live deployment.

3.3 Portable command-line build guidance

If using a non-Visual Studio toolchain, compile QSC first and then AERN. The exact file list depends on the configured algorithm set. For validation, avoid AVX-specific files unless the target machine and compiler configuration are known to support them.

```

# Example outline only; adjust source list and platform flags to match the deployment.
cc -I./QSC -c QSC/*.c
cc -I./QSC -I./AERN -c AERN/*.c
cc -I./QSC -I./AERN -I./AERNTTest -c AERNTTest/*.c
cc -o AERNTTest *.o

```

Build rule: The operational manual assumes the AERN library and AERNTTest validation harness build successfully and pass the

complete test sequence. If a deployment changes cryptographic parameter sets, run the complete test harness again before issuing certificates.

4. Certificate and key material lifecycle

AERN separates identity keys from tunnel keys. Certificates authenticate devices and administrative messages. Tunnel keys are derived from authenticated KEM exchanges and are scoped to a peer and direction. A signing-key compromise affects future authentication; it does not by itself compute completed tunnel state after ephemeral KEM material and active symmetric state have been erased.

4.1 Certificate classes

Certificate	Holder	Created by	Signed by	Purpose
Root certificate	ARS and all domain participants	ARS	Self/root root process	Trust anchor used to validate child certificates.
ADC child certificate	ADC and optionally ARS remote-signing service	ADC	ARS	Authorizes domain control and remote signing proxy operations.
APS child certificate	Each proxy server	APS	ARS directly or ARS through ADC proxy signing	Authorizes proxy mesh participation.
Client child certificate	Each client device	Client	ARS directly or through ADC proxy signing	Authorizes client registration and entry-node connection.

4.2 Certificate validity policy

The default certificate validity period is one year. The maximum configured certificate period is twice the default period, and the minimum period is one day. A child certificate must not exceed the root certificate expiration time. When the root certificate expires, every child certificate must be renewed under the new root.

Parameter	Value	Operational effect
AERN_CERTIFICATE_DEFAULT_PERIOD	365 days	Default requested certificate duration.
AERN_CERTIFICATE_MAXIMUM_PERIOD	730 days	Maximum configured certificate period.
AERN_CERTIFICATE_MINIMUM_PERIOD	1 day	Minimum configured certificate period.
AERN_MFK_EXPIRATION_PERIOD	60 days	Nominal peer tunnel key validity period before renewal policy should rekey.

4.3 Key custody rules

Material	Custody requirement
ARS signing key	Keep offline or on an isolated root host. Use network exposure only for narrowly controlled remote signing.
ADC signing key	Protect as a domain-control key. Compromise permits topology and administrative-message abuse.
APS signing key	Protect on each proxy. Compromise permits identity abuse by that proxy.
Client signing key	Protect on the client host or secure token where available.
KEM ephemeral secret	Erase after exchange completes.
RCS tunnel state	Erase on tunnel teardown, peer revocation, key expiration, or rekey.

Log seed/salt

Store only as required by the encrypted log mechanism; protect log passwords.

5. Command modes and command reference

The current AERN application exposes role-specific interactive command modes. Exact executable names depend on the build package. The command examples below use role names as binary placeholders: aern-ars, aern-adc, aern-aps, and aern-client.

5.1 Common console modes

Mode	Prompt purpose	Common commands
user	Initial command mode.	enable, help, quit
enable	Privileged operational mode.	config, show config, show log, clear, exit, quit
config	Host and service configuration mode.	address, crypto, name domain, name host, retries, timeout, log, server, clear, exit
certificate	Certificate generation, import/export, and printing.	generate, export, import, print, sign on ARS, revoke on ADC where supported
server	Network service, topology, registration, and administrative actions.	service start stop pause resume, register, resign, list, announce, converge, revoke, sproxy, connect, query

5.2 Common configuration commands

Command	Applies to	Purpose	Example
enable	All roles	Enter privileged mode after authentication.	enable
config	All roles	Enter configuration mode.	config
address [ip-address]	All server roles	Set local network interface address; restart is required for a running service.	address 10.10.0.20
name domain [domain-name]	All roles	Set domain name.	name domain aern.example.net
name host [host-name]	All roles	Set canonical host name.	name host aps-01
timeout [seconds]	All roles	Set default session or console timeout.	timeout 120
retries [count]	All roles	Set authentication retry limit.	retries 2
log enable disable	All roles	Enable or disable administrative logging.	log enable
crypto	All roles	Enter certificate/cryptographic command interface.	crypto
server	All roles	Enter server/network command mode.	server

5.3 Certificate-mode commands

Command	Role	Purpose
generate [period]	ARS, ADC, APS, Client	Generate local signing key pair and local certificate request or root certificate, depending on role. Period is the requested validity interval.
export [path]	All roles	Export the local certificate or certificate request to a file for signing or transfer.
import [path]	All roles	Import a signed certificate, root certificate, ADC certificate, or peer certificate according to the role workflow.

print	All roles	Print local certificate information.
sign [certificate path]	ARS	Sign a child certificate directly with the ARS signing key.
revoke [certificate path]	ADC	Revoke a certificate where ADC certificate-revoke mode is supported.
exit	All roles	Return to config mode.

5.4 Server-mode and ADC console commands

Command	Role	Purpose
service start	ARS, ADC, APS	Start the network service after certificate and configuration prerequisites are satisfied.
service stop	ARS, ADC, APS	Stop the service.
service pause	ARS, ADC, APS	Pause service processing where supported.
service resume	ARS, ADC, APS	Resume a paused service.
register [address]	APS, Client	Register with the ADC or selected remote service.
resign [address]	APS, Client	Resign from the ADC-controlled topology.
list	ADC, APS, Client	Display local topology or known nodes.
announce [certificate-path, ip-address]	ADC	Announce a certified APS to the network.
converge	ADC	Trigger topology convergence over APS nodes.
revoke [certificate path]	ADC	Broadcast revocation for an APS certificate.
sproxy [certificate path]	ADC	Send a remote signing request to the ARS service.
connect [ip address]	Client	Connect to an application server or client through AERN where supported.
query [canonical-name]	Client	Request node information from ADC.
topology	ADC administrative console	Print all topology nodes.
certificate [serial]	ADC administrative console	Print certificate details for a hex serial number.
log [last N]	ADC administrative console	Print recent administrative log entries.
status	ADC administrative console	Print uptime and connection status.

6. ARS root initialization

The ARS is the first node initialized. It creates the root certificate and signs child certificates. In a high-assurance deployment, the ARS should be offline except during controlled certificate operations. If remote signing is enabled, only the ADC should be able to request signing operations.

6.1 ARS first-run checklist

Step	Action	Acceptance criterion
1	Install AERN and QSC on the isolated ARS host.	Executable starts and can enter enable/config mode.
2	Set host name, domain name, and ARS network address.	Configuration prints expected identity.

3	Generate root certificate with selected validity period.	Root certificate prints with correct issuer, serial, algorithm, version, and expiration.
4	Export root certificate to the domain distribution path.	Every ADC, APS, and client can import the same root certificate.
5	Sign ADC child certificate.	ADC certificate validates against root.
6	Start remote-signing service only if required.	Service is reachable only from ADC or management VLAN.

6.2 ARS command transcript

Start the ARS application. Executable name is deployment-specific.
aern-ars

```
enable
config
name domain aern.example.net
name host ars-root-01
address 10.10.0.10
log enable
crypto
generate 31536000
print
export C:\AERN\certs\ars-root.cert
exit
server
service start
```

6.3 Direct signing of the ADC certificate

On ADC, generate and export the unsigned ADC certificate request.

```
aern-adc
enable
config
name domain aern.example.net
name host adc-01
address 10.10.0.20
crypto
generate 31536000
export C:\AERN\certs\adc-01.request
```

Transfer adc-01.request to the ARS signing environment.

```
# On ARS:
crypto
sign C:\AERN\certs\adc-01.request
export C:\AERN\certs\adc-01.signed.cert
```

Transfer adc-01.signed.cert back to ADC.

```
# On ADC:
crypto
import C:\AERN\certs\adc-01.signed.cert
print
```

Signing discipline: The ARS signing key must never be copied to the ADC, APS, or client. The certificate request moves to the ARS; only the signed certificate returns.

7. ADC initialization and remote signing channel

The ADC manages domain registration, topology, revocation, convergence, and administrative messages. The ADC must hold a valid ARS-signed certificate before it performs domain-control operations.

7.1 ADC initialization sequence

```
aern-adc

enable
config
name domain aern.example.net
name host adc-01
address 10.10.0.20
log enable
crypto
import C:\AERN\certs\ars-root.cert
import C:\AERN\certs\adc-01.signed.cert
print
exit
server
service start
status
```

7.2 Enabling remote signing

Remote signing allows the ADC to forward a device certificate request to the isolated ARS. This reduces the need to expose the ARS to every network device. The ARS must have the signed ADC certificate installed as the only authorized remote signing requester.

```
# On ARS: import the signed ADC certificate for remote-signing authorization.
aern-ars
enable
config
crypto
import C:\AERN\certs\adc-01.signed.cert
exit
server
service start

# On ADC: submit a device certificate request to ARS through the signing proxy.
aern-adc
enable
config
server
sproxy C:\AERN\certs\aps-01.request
```

7.3 ADC administrative operations

Operation	Command	State changed	When to use
Register device	register [address] or register/update message path	Topology and certificate cache	When APS or client joins the domain.
Announce APS	announce [certificate-path, ip-address]	Topology update broadcast	When an APS is added or refreshed.
Converge topology	converge	Topology validation and update messages	When nodes need synchronized topology/certificate confirmation.
Revoke certificate	revoke [certificate path] or revoke [serial]	Topology removal and peer state teardown	When a node is retired, expired, failed, or administratively removed.
Remote sign	sproxy [certificate path]	ARS-signed certificate returned through ADC	When ARS is isolated and ADC acts as signing proxy.
Query topology	topology, certificate [serial], status	No relay state change	Routine operations and auditing.

8. APS proxy initialization

An APS proxy server participates in the full relay mesh. It must possess an ARS-signed certificate, register with the ADC, synchronize topology, obtain peer certificates, and establish APS-to-APS encrypted tunnels. The APS service should not be started for relay traffic until those prerequisites are complete.

8.1 APS first-run command transcript

```
aern-aps

enable
config
name domain aern.example.net
name host aps-01
address 10.10.1.11
log enable
crypto
import C:\AERN\certs\ars-root.cert
generate 31536000
export C:\AERN\certs\aps-01.request

# Sign using ARS direct signing or ADC remote signing.
# After signing returns:
import C:\AERN\certs\aps-01.signed.cert
print
exit
server
register 10.10.0.20
list
service start
```

8.2 APS mesh synchronization

After registration, the APS obtains a topology list from the ADC. APS nodes are sorted consistently by serial number. Route hints are one-byte, one-based APS topology ordinals; zero is reserved as terminator and empty value. The active compact route format supports at most 255 route-addressable APS entries in one route domain.

Synchronization step	Message family	Result
Register APS with ADC	register_request/register_response or register_update variants	ADC validates certificate and records topology node.
Obtain topology	register_update_response or join_response	APS obtains ordered APS list.
Fetch peer certificates	incremental_update_request/response	APS validates peer certificate against topology hash and root signature.
Establish peer tunnel	mfk_exchange_request/response	APS derives directional transmit and receive RCS states.
Validate topology freshness	converge_request/response/update	Topology remains synchronized and stale entries are detected.

8.3 APS service-state checks

Check	Expected value
Signed certificate installed	Certificate chains to ARS root and has APS designation.
Topology count	At least three route-addressable APS nodes for full relay operation.
Peer certificates	Every peer APS certificate validates against topology entry.

Connection states	Each APS peer has initialized transmit and receive cipher state.
Replay policy	rxseq and txseq start at defined initial values; replay window is zero; strict sequencing enabled.
Backend callback	Egress/ingress callback installed before live traffic.

9. Client initialization and entry-node connection

A client is an authenticated endpoint that obtains the current APS topology from the ADC and establishes an encrypted tunnel to a selected entry APS. Proxy servers do not retain general client certificates as topology members in the same way they retain APS peers; clients use ADC registration/join operations to obtain current proxy lists.

9.1 Client first-run command transcript

```
aern-client

enable
config
name domain aern.example.net
name host client-01
address 0.0.0.0
log enable
crypto
import C:\AERN\certs\ars-root.cert
generate 31536000
export C:\AERN\certs\client-01.request

# Sign using direct ARS signing or ADC remote signing.
import C:\AERN\certs\client-01.signed.cert
print
exit
server
register 10.10.0.20
list
connect 10.10.1.11
```

9.2 Entry-node selection

The client selects an entry APS from the topology received from ADC. A production client may use random selection, policy-based selection, or latency-aware selection, provided it does not introduce persistent metadata correlation beyond the intended operational policy. The entry tunnel is established using the authenticated KEM exchange and derives directional symmetric state.

10. Network synchronization and administrative messages

Administrative message processing is the control plane of AERN. It handles announcement, convergence, registration, certificate retrieval, remote signing, revocation, resignation, topology query, and topology status. These messages are signed and carry freshness state where implemented. They should be tested in every deployment before live relay traffic is enabled.

Message family	Initiator	Receiver	Operational effect
announce_broadcast/response	ADC	APS/Client population	Announces or refreshes a topology node.
converge_request/response	ADC	APS	Confirms or corrects topology node state.
converge_update	APS	ADC	Submits APS-signed topology correction.
incremental_update_request/response	APS or Client	Peer node	Retrieves and verifies a peer certificate.
fkey_request/response	APS	APS	Administrative fragment-key operation; not MPDC multi-party strengthening.

mfk_exchange_request/response	APS or Client	APS or peer	Authenticated KEM-based tunnel establishment.
register_request/response	Client or APS	ADC	Basic certificate-authenticated registration contact.
register_update_request/response	Client or APS	ADC	Registration with topology-list transfer.
register_update_v2_response	ADC	Client or APS	Versioned topology-list response.
remote_signing_request/response	ADC	ARS	ARS signs device certificate through ADC proxy.
resign_request/response	Client or APS	ADC	Device leaves the network; ADC may broadcast revocation.
revoke_broadcast/response	ADC	APS population	Remove certificate and topology state for revoked node.
join_request/response	Client	ADC	Checks topology hash and obtains updates.
topological_query_request/response	Client or APS	ADC	Fetches authenticated topology node by query.
topological_status_request/response	ADC/APS/Client	Target node	Signed liveness/status check.

10.1 Administrative test policy

Administrative tests should validate current-design behavior: valid path, malformed/truncated parser rejection, corrupted payload/signature rejection, stale timestamp rejection where timestamp checking is implemented, replay or bad sequence rejection where sequence checking is implemented, and no unintended mutation of topology, certificate cache, tunnel state, session state, or queues after rejection.

Administrative trust model: AERN assumes competent administration of ARS and ADC. Do not add speculative tests that assume a malicious administrator unless the implementation explicitly enforces that policy.

11. Relay packet format and tunnel behavior

The active relay wire packet is a fixed 1500-byte packet. The visible outer header is 22 bytes and is authenticated as associated data. The 1478-byte ciphertext region contains the authenticated encrypted relay plaintext and authentication tag. After authentication and decryption, the relay plaintext is 1446 bytes.

Component	Size	Visibility	Operational note
Relay wire packet	1500 bytes	Wire-visible length	Fixed MTU relay transmission unit.
Outer tunnel header	22 bytes	Visible but authenticated	Contains flag, ciphertext length, sequence number, UTC timestamp, and reserved byte.
Ciphertext region	1478 bytes	Opaque	Authenticated encrypted tunnel payload and tag.
Relay plaintext	1446 bytes	Visible only after tunnel authentication	Contains length prefix, route path, relay header, body, and padding.
Length prefix	2 bytes	Encrypted	Used relay payload length after route path.
Route path	16 bytes	Encrypted	Consumed-next-hop one-byte route hints.
Relay payload header	32 bytes	Encrypted	Session, packet, fragment, payload type, reserved byte, and flags.
Data payload per fragment	1396 bytes	Encrypted	Maximum backend payload bytes per relay fragment.

11.1 Tunnel state policy

State	Policy
txseq	Increment only after successful send/encryption. Prevent wrap at UINT64_MAX.
rxseq	Increment only after freshness, sequence, and authentication acceptance.
authfail	Increment on authentication failure; success resets according to tunnel policy.
RCS state after auth failure	Do not assume the RCS internal state remains usable. Treat repeated failures as tunnel risk and rekey or tear down at threshold.
replay window	Zero; strict next-sequence processing is used for ordered APS tunnels.
timestamp threshold	Default 60 seconds. Packets outside threshold are rejected.

12. Session-open, backend transport, and client message flow

AERN uses explicit ingress-to-egress relay sessions. The ingress creates a pending session when a new outbound flow is submitted. It sends a session-open payload to the selected egress and queues non-dummy data until a valid session-open acknowledgement is received. The egress creates active egress-side session state only after validating the session-open payload.

12.1 Session-open structures

Structure	Size	Fields
session-open	36 bytes	sessionid, 22-byte destination address, ingress hint, egress hint, destination port, reserved byte, flags
session-open-ack	12 bytes	sessionid, status, flags, reserved 16-bit field
relay payload header	32 bytes	sessionid, packetid, fragseq, fragcount, msglen, payloadtype, reserved, flags

12.2 Example HTTPS-like opaque client flow

```

Client -> Entry APS:
  Serialized destination: 203.0.113.10:443
  Opaque payload: TLS ClientHello-like byte string

Entry APS:
  Creates pending(sessionid)
  Builds session-open to selected egress APS
  Queues outbound payload

Egress APS:
  Authenticates packet and route path
  Validates session-open
  Creates active egress session
  Returns session-open-ack

Entry APS:
  Accepts acknowledgement
  Marks ingress session active
  Releases queued payload

Egress backend:
  Receives opaque serialized payload exactly once
  Returns opaque server response through return interface

Ingress callback:
  Receives return payload with return flag validated

```

12.3 Backend boundary requirements

The backend adapter may bind AERN to a TUN or TAP device, raw socket path, userspace TCP/UDP stack, packet filter, or application adapter. It is outside the cryptographic relay core. It must receive only authenticated, session-valid, reassembled serialized packets. It must not log relay route paths, session identifiers, packet identifiers, fragment identifiers, or client payload contents in ordinary operational logs.

13. Fragmentation and reassembly operations

Fragmentation occurs inside the encrypted relay payload layer. The outer packet header does not expose fragment metadata. Each fragment carries the same session identifier and packet identifier, while fragment sequence, fragment count, length, type, reserved byte, and direction flag are inside the encrypted relay payload header.

Parameter	Value	Operational effect
AERN_RELAY_DATA_PAYLOAD_SIZE / AERN_FRAG_CHUNK_SIZE	1396 bytes	Maximum backend payload bytes in one relay fragment.
AERN_MAX_FRAGMENTS	4096	Maximum fragment count for one fragmented message.
AERN_RELAY_FRAGMENT_CACHE_TIMEOUT_MILLISECONDS	30000 ms	Incomplete fragment sets expire after this timeout.
Fragment ordering	Reorderable after tunnel authentication	Tunnel packets are strictly sequenced; fragment cache may accept out-of-order fragment sets after authentication.

13.1 Fragment acceptance checklist

Validation	Requirement
Tunnel authentication	The enclosing tunnel packet must authenticate and decrypt successfully before fragment cache insertion.
Session binding	Session identifier and context must match an active session.
Packet binding	All fragments in one set use the same packet identifier.
Fragment count	Count must be within configured maximum and consistent across fragments.
Fragment lengths	Non-final fragments must be full-sized; final fragment may be short.
Direction	Outbound and return fragments must not mix.
Payload type/reserved byte	Payload type and reserved byte must remain consistent.
Delivery	Backend callback is invoked exactly once after complete reassembly.

14. Dummy traffic, ingress delay, and traffic shaping

AERN dummy traffic and ingress delay are traffic-analysis mitigations. They are local policy controls and are not a complete defense against a fully global timing adversary. Dummy packets use the same fixed packet size and tunnel encryption as ordinary relay packets. Terminal APS nodes discard authenticated dummy payloads without backend delivery.

Parameter	Default	Tuning guidance
AERN_DUMMY_TRAFFIC_ENABLED	1	Disable only for isolated lab debugging; enable for realistic sparse-flow operation.
AERN_DUMMY_TRAFFIC_BANDWIDTH_FLOOR_PERCENT	10%	Below this local utilization, dummy generation is preferred. Higher values increase cover traffic.
AERN_DUMMY_TRAFFIC_BANDWIDTH_CEILING_PERCENT	25%	At or above this utilization, dummy generation is suppressed. Lower values reduce overhead.

AERN_DUMMY_TRAFFIC_INTERVAL_MINIMUM_MILLISECONDS	50 ms	Minimum generation interval. Lower values increase CPU/network use.
AERN_DUMMY_TRAFFIC_INTERVAL_MAXIMUM_MILLISECONDS	250 ms	Maximum generation interval. Higher values reduce cover density.
AERN_DUMMY_TRAFFIC_WINDOW_TARGET_BYTES	128 MTU packets	Local utilization denominator. Tune with expected APS throughput.
AERN_DUMMY_TRAFFIC_MAXIMUM_PER_WINDOW	8	Caps dummy emission per accounting window.
AERN_RELAY_TRAFFIC_WINDOW_MILLISECONDS	1000 ms	Traffic accounting reset window.
AERN_INGRESS_DELAY_ENABLED	1	Enable for modest timing obfuscation.
AERN_INGRESS_DELAY_MINIMUM_MILLISECONDS	0 ms	Lower delay bound.
AERN_INGRESS_DELAY_MAXIMUM_MILLISECONDS	25 ms	Upper delay bound; increasing may affect latency-sensitive flows.

15. Replay, freshness, failure handling, and logging

15.1 Replay and freshness

AERN uses strict next-sequence replay protection. A packet is accepted only if the authenticated sequence number equals the next expected receive sequence and the timestamp lies within the valid-time threshold. Duplicate packets, old packets, future sequence packets, stale timestamps, modified timestamps, and modified sequence numbers are rejected. The sequence and timestamp are authenticated as associated data in the tunnel layer.

15.2 Failure handling

Failure	Expected behavior
Malformed outer header	Reject before relay-state mutation.
Bad timestamp	Reject without advancing rxseq.
Unexpected sequence	Reject without advancing rxseq.
Authentication failure	Reject, increment authfail, and do not advance AERN rxseq. RCS internal state may require tunnel renewal.
Authentication threshold	Return channel-down behavior and tear down or renew according to local policy.
Missing next-hop tunnel	Return key-unrecognized or channel-down according to failure location; do not call backend.
Backend callback failure	Propagate failure while preserving session integrity unless local policy tears down the session.
Expired session/fragment/queue item	Cleanup removes expired state.

15.3 Logging discipline

Operational logging is for device, topology, service, and error administration. Logs may record device identity, event time, certificate state, topology state, and error class. Logs must not record client content, external destination payloads, route paths, relay session identifiers, packet identifiers, fragment identifiers, or backend payload bytes.

16. Parameter tuning reference

Tuning domain	Primary parameters	Recommended approach
Certificate validity	AERN_CERTIFICATE_DEFAULT_PERIOD, MINIMUM, MAXIMUM	Use shorter ADC and APS certificates for high-security deployments; ensure root lifetime exceeds all children.
Route length	AERN_ROUTE_MINIMUM_HOPS, AERN_ROUTE_MAXIMUM_HOPS, route path capacity	Increase hops only when latency budget allows. Keep route-addressable APS set within 255 for compact profile.

Replay/freshness	AERN_RELAY_STRICT_SEQUENCING, AERN_RELAY_REPLAY_WINDOW_SIZE, AERN_PACKET_TIME_THRESHOLD	Keep strict sequencing and zero replay window unless transport semantics change. Tune timestamp threshold to clock synchronization quality.
Fragments	AERN_FRAG_CHUNK_SIZE, AERN_MAX_FRAGMENTS, fragment timeout	Large fragment counts increase memory pressure. Tune timeout for expected jitter and backend packet size.
Sessions	Session-open timeout, active/idle expiry	Shorter pending timeouts reduce queue buildup; longer active timeouts reduce repeated session-open overhead.
Dummy traffic	Floor, ceiling, interval, window, max per window	Tune from observed traffic patterns. Avoid cover-traffic settings that saturate links.
Ingress delay	Min/max delay	Keep small for interactive traffic; increase only where latency budget allows.
Logging	Log enable, encrypted log parameters	Enable for operations; ensure metadata discipline and protected log storage.

17. Operational runbooks

17.1 Bring-up runbook

Order	Operation	Stop condition
1	Build QSC, AERN, and AERNTTest.	Stop if the test executable fails.
2	Initialize ARS and export root certificate.	Stop if root certificate does not self-validate.
3	Generate and sign ADC certificate.	Stop if ADC certificate does not validate against root.
4	Start ADC service.	Stop if ADC cannot enter service state.
5	Generate APS certificates and sign them through ARS or ADC sproxy.	Stop if any APS certificate fails root validation.
6	Register APS nodes with ADC.	Stop if topology is unsorted, stale, or inconsistent.
7	Run convergence and incremental certificate synchronization.	Stop if any peer certificate hash mismatch occurs.
8	Establish APS-to-APS MFK/MEK tunnels.	Stop if directional state roundtrip fails.
9	Initialize and register client.	Stop if client cannot obtain topology.
10	Establish client-to-entry tunnel and run test payload.	Stop if backend callback is not invoked exactly once for valid data.

17.2 Revocation runbook

```
# On ADC administrative console or server mode:
revoke <certificate-path-or-serial>
converge
topology
log last 20
```

Expected result:

- revoked node removed from active topology;
- peer tunnel state destroyed or made unusable;
- relay sessions involving the revoked peer cleared;
- clients receive refreshed topology at next join/update;
- logs contain device/error information only.

17.3 Rekey and renewal runbook

1. Identify peer tunnel state approaching MFK expiration.
2. Suspend or queue affected relay traffic if the deployment policy requires it.
3. Execute authenticated MFK/MEK exchange.
4. Verify transmit-to-receive and receive-to-transmit direction agreement.
5. Reset or reinitialize txseq, rxseq, authfail, and cipher state according to tunnel policy.
6. Erase superseded key-dependent state.
7. Resume relay traffic.

18. Validation checklist and troubleshooting

Validation area	Required tests
Certificate lifecycle	Root creation, child signing, import/export roundtrip, expiration bounds, root serial binding.
ADC messages	All administrative message families, valid paths, malformed inputs, bad signatures, stale timestamps, replay where implemented.
Tunnel establishment	APS-to-APS and client-to-APS MFK/MEK exchange, directional key agreement, nonce separation, sequence initialization.
Packet header	22-byte header roundtrip, reserved byte, AAD mutation rejection, fixed 1500-byte relay packet model.
Routes	One-byte hints, zero terminator, no serialized cursor, route generation, consumption, invalid hint rejection.
Sessions	Session-open, session-open-ack, pending queue, active session, duplicate open, timeout cleanup.
Payload header	32-byte layout, payload types, return flag, reserved byte, invalid header rejection.
Fragments	Boundary, multi-fragment, out-of-order, duplicate, altered duplicate, timeout, backend exactly once.
Backend	Egress and ingress callback boundaries, wrong direction rejection, callback failure propagation.
Dummy/delay	Traffic policy bounds, dummy discard, ingress delay flush.
Replay/freshness	Strict sequence, duplicate, future/old sequence, timestamp failures, no rxseq mutation on failure.
Logging	No relay-sensitive metadata in ordinary logs.

18.1 Troubleshooting table

Symptom	Likely cause	Resolution
ADC will not start service	Unsigned or invalid ADC certificate.	Import root and signed ADC certificate; verify certificate print output.
APS registration rejected	Certificate root signature failure or topology mismatch.	Regenerate APS request, sign through ARS/ADC, re-register.
Peer tunnel fails	MFK/MEK signature, timestamp, or KEM failure.	Check clock synchronization, certificate chain, algorithm set, and key-exchange logs.
Relay packet rejected as replay	Unexpected rxseq or duplicate packet.	Verify ordered transport, no packet replay, and sequence reset after rekey.
Backend not called	Session not active, route not terminal, dummy/error payload, wrong direction, or backend callback missing.	Inspect session-open ack, route terminal status, payload type, and callback installation.
Fragment never delivered	Missing fragment, inconsistent metadata, timeout, or wrong direction.	Check fragment count, sequence, packet ID, session ID, and timeout.
High dummy overhead	Floor too high, ceiling too high, interval too low, max per window too high.	Reduce dummy floor, reduce maximum per window, or increase interval.
Timing correlation risk	Low traffic, no dummy traffic, no ingress delay.	Enable dummy traffic and ingress delay; increase cover traffic within bandwidth budget.

Appendix A. Message family reference

This appendix lists each operational message family, its normal initiator, payload purpose, and implementation notes. It is intended as an operator-facing summary rather than a wire-format replacement for the source headers.

Family	Initiator	Payload purpose	Operator note
announce	ADC	Topology node and optional certificate material	Use to distribute new or refreshed node information.
converge	ADC / APS	Topology node confirmation or correction	Use after suspected topology drift or certificate refresh.
incremental update	APS or Client	Certificate serial query and certificate response	Used by nodes to fetch peer certificates after topology updates.
fkey	APS	Token-bound administrative fragment-key material	Active API operation; not MPDC multi-party fragment strengthening.
mfk/mek exchange	APS or Client	Certificate-bound KEM exchange messages	Creates directional tunnel state.
register	APS or Client	Device certificate and ADC certificate response	Basic enrollment contact.
register update	APS or Client / ADC	Device certificate, topology list, ADC certificate	Enrollment with topology transfer.
remote signing	ADC / ARS	Unsigned child certificate and signed child certificate	Only ADC should request ARS remote signing.
resign	APS or Client / ADC	Serial-bound resignation request	Used when leaving domain.
revoke	ADC	Serial or certificate-bound revocation	Removes node from topology and clears associated state.
join	Client / ADC	Client topology hash and updated topology list if needed	Used before client selects an entry APS.
topological query	Client or APS / ADC	Issuer or serial query and node response	Used to locate and validate a node.
topological status	ADC/APS/Client	Signed liveness/status challenge and response	Used for maintenance and availability checks.

Appendix B. Example deployment transcript

```
# ARS
start aern-ars
enable
config
name domain aern.example.net
name host ars-root-01
address 10.10.0.10
crypto
generate 31536000
export C:\AERN\certs\ars-root.cert
server
service start

# ADC
start aern-adc
enable
config
name domain aern.example.net
name host adc-01
address 10.10.0.20
crypto
import C:\AERN\certs\ars-root.cert
```

```

generate 31536000
export C:\AERN\certs\adc-01.request

# ARS signs ADC
crypto
sign C:\AERN\certs\adc-01.request
export C:\AERN\certs\adc-01.signed.cert

# ADC imports signed certificate and starts
crypto
import C:\AERN\certs\adc-01.signed.cert
server
service start

# APS nodes repeat with unique host/address values
start aern-aps
enable
config
name domain aern.example.net
name host aps-01
address 10.10.1.11
crypto
import C:\AERN\certs\ars-root.cert
generate 31536000
export C:\AERN\certs\aps-01.request

# ADC remote signing through ARS
server
sproxy C:\AERN\certs\aps-01.request
crypto
import C:\AERN\certs\aps-01.signed.cert
server
register 10.10.0.20
service start

# Client
start aern-client
enable
config
name domain aern.example.net
name host client-01
crypto
import C:\AERN\certs\ars-root.cert
generate 31536000
export C:\AERN\certs\client-01.request
# sign and import client-01.signed.cert
server
register 10.10.0.20
connect 10.10.1.11

```

Appendix C. Operator cautions

Caution	Reason
Do not allocate AERN_MESSAGE_MAX_SIZE on the stack.	The maximum message size is sized for worst-case administrative/key-exchange material, especially large KEM public keys. Use exact operation sizes or heap allocation.
Do not assume RCS remains synchronized after authentication failure.	AERN rxseq does not advance before authentication success, but internal RCS state may be invalid after failure. Rekey or tear down at policy threshold.
Do not treat IDG structures as active route behavior.	IDG fields are reserved for future inter-domain gateway profiles.
Do not expose backend payload or relay metadata in logs.	Logging sensitive metadata can defeat the privacy properties of the relay layer.

Do not exceed 255 active route-addressable APS hints in the compact route profile.

Route hints are one byte and zero is reserved. Larger deployments need route domains, shards, or an extended-hint profile.

Appendix D. Deployment integration profile

This appendix defines a deterministic deployment checklist from certificate creation through relay validation. It is intended for administrators who are implementing a working AERN domain. The command names remain role placeholders where local deployments use different service wrappers.

D.1 Required deployment order

Phase	Operator action	Acceptance criterion
1. Build	Build QSC, AERN, and AERNTTest with the same algorithm profile and platform settings.	The AERN library links successfully and the test harness starts.
2. ARS	Generate the ARS root certificate and protect the ARS signing key.	The root certificate prints with the expected issuer, serial, algorithm, version, and validity interval.
3. ADC	Generate the ADC child certificate request, sign it with ARS, import the signed ADC certificate, and start the ADC service.	The ADC certificate validates against the root and the ADC administrative console reports a usable topology version.
4. Remote signing	Where used, install the signed ADC certificate on the ARS remote-signing service and restrict network access to the ADC.	A test device certificate request can be signed through the ADC proxy path and rejected when not ADC-authorized.
5. APS certification	Generate each APS certificate request, sign it directly or through ADC remote signing, and import root and signed child certificates.	Each APS certificate validates against the ARS root and has APS designation.
6. APS registration	Register each APS with the ADC and verify the sorted topology list.	The ADC topology count and version increase monotonically; stale update attempts are rejected.
7. APS mesh synchronization	Run incremental certificate retrieval and MFK/MEK tunnel establishment between APS peers.	Every route-eligible APS peer reaches synchronized mesh status and has initialized transmit and receive cipher state.
8. Client enrollment	Generate, sign, and import the client certificate, then register or join with the ADC.	The client receives a valid APS topology list and can select an entry APS.
9. Entry tunnel	Establish the client-to-entry APS tunnel.	The entry tunnel reaches active state and rejects replay, stale timestamp, and authentication-failure tests.
10. Relay validation	Run session-open, pending queue, opaque delivery, return delivery, fragmentation, dummy traffic, ingress delay, and teardown validation.	Backend callbacks receive only authenticated, session-valid, non-dummy, reassembled serialized packets.

D.2 Administrator validation commands and observations

AERN deployments expose role command modes and administrative helpers rather than a single universal deployment executable. Administrators should bind the following checks to the local wrapper or service manager used in the deployment package.

Check	Expected observation	Failure response
show config / print certificate	Issuer, host, domain, algorithm set, version, serial, and validity interval match the deployment plan.	Stop enrollment and regenerate the certificate or configuration before signing.
ADC topology display	Topology entries are sorted consistently, APS designations are correct, and the topology version is nonzero.	Reject relay startup until registration and version state are corrected.
APS list or peer status	Each required peer is synchronized, not expired, not failed, and has a usable connection state.	Remove the peer from route eligibility and repeat certificate retrieval plus MFK/MEK exchange.
Replay test	Duplicate, stale, future-sequence, and modified-header packets are rejected without rxseq advancement.	Treat as a release-blocking defect in tunnel validation.
Session-open test	Pending data remains queued until a valid session-open acknowledgement is accepted.	Disable backend delivery until session-state handling is corrected.
Fragment test	Out-of-order fragments reassemble only after all authenticated fragments arrive and metadata is consistent.	Clear fragment cache and investigate bounds, sequence, direction, and length handling.
Dummy traffic test	Authenticated dummy packets are discarded silently and never delivered to a backend.	Treat backend delivery of dummy payloads as a release-blocking defect.
Backend callback test	Callbacks receive only serialized payloads, not route paths, session cache internals, or fragment metadata.	Revise the adapter boundary before production use.

D.3 Production hardening policy

- Keep the ARS offline whenever direct certificate signing is not required. When remote signing is enabled, restrict ARS network reachability to the ADC management interface or management VLAN.
- Protect the ADC signing key as a domain-control key. ADC compromise can bias topology, suppress revocation, and degrade anonymity even when it cannot directly decrypt completed tunnel traffic.
- Use NTP or an authenticated time source for ARS, ADC, and APS hosts. The default packet time threshold is 60 seconds, and incorrect clocks will cause valid packets or administrative messages to be rejected.
- Keep strict sequencing enabled for APS tunnels. The current replay-window size is zero and the ordered tunnel model assumes exact next-sequence acceptance.
- Do not run a production relay domain with fewer than three route-addressable APS nodes. A practical validation network should use at least four APS nodes to exercise interior route variation.
- Treat the compact route profile as limited to 255 route-addressable APS ordinals. Larger physical deployments should use sharding or a separately specified extended-hint profile.
- Bound pending queues, delay queues, fragment caches, and backend work queues. Queue exhaustion is an operational availability risk even when cryptographic authentication remains intact.
- Do not log route paths, relay session identifiers, relay packet identifiers, fragment identifiers, backend payload bytes, dummy packet bodies, or client content. Keep operational logs restricted to device identity, certificate status, topology state, service state, and error class.
- Run AERNTTest and a virtual-network validation after any change to constants, packet sizes, algorithm sets, replay policy, route handling, session handling, fragment handling, dummy policy, or backend adapters.

D.4 Minimal virtual-network validation transcript

A validation harness should instantiate one ARS, one ADC, at least four APS nodes, one client, and one backend callback endpoint. The harness should run the following sequence before the domain is considered deployable:

- Generate ARS root certificate and import it into ADC, APS, and client contexts.
- Generate and root-sign ADC, APS, and client certificates.
- Start ADC and confirm topology version begins at a nonzero initialized value after first topology mutation.
- Register each APS and verify that topology count, sorted order, and version increment as expected.
- Perform incremental certificate retrieval for every APS peer and compare certificate hashes to topology entries.
- Execute MFK/MEK exchanges between APS peers and verify synchronized peer status.
- Register or join the client and verify that it receives the APS topology list.
- Establish the client-to-entry APS tunnel and reject duplicate, stale, malformed, and authentication-failing packets.
- Submit an opaque HTTPS-like serialized packet, open an ingress-to-egress relay session, hold data pending until acknowledgement, and deliver the payload to the egress backend exactly once.
- Inject return traffic through the egress return interface and verify terminal delivery at the ingress callback with the return flag validated.
- Send an oversized serialized packet and verify encrypted relay fragmentation, out-of-order reassembly, timeout cleanup, and no duplicate backend delivery.
- Enable dummy generation and verify dummy discard at terminal APS without backend delivery.
- Revoke one APS and verify route eligibility removal, tunnel teardown, cache cleanup, and stale topology rejection.

D.5 Tuning recommendations by deployment class

Deployment class	Recommended starting policy	Operational note
Laboratory validation	Four APS nodes, strict sequencing, dummy enabled, ingress delay enabled, default fragment and session timeouts.	Exercise all state transitions before reducing instrumentation.
Low-latency enterprise VPN domain	Three to eight APS nodes per active route domain, route maximum adjusted by latency budget, dummy floor kept modest.	Prioritize predictable latency while preserving fixed-size packets and encrypted route metadata.
High-assurance institutional domain	Shorter ADC/APS certificate lifetimes, isolated ARS, remote signing restricted to ADC, frequent topology convergence checks.	Administrative hardening is more important than increasing hop count.
Large proxy estate	Partition APS nodes into route-addressable shards of at most 255 ordinals or define an extended-hint profile.	The compact route path is not a million-node addressing scheme despite larger theoretical topology constants.
Sparse traffic domain	Keep dummy traffic enabled and tune floor, ceiling, and per-window cap from observed utilization.	Dummy traffic reduces sparse-flow observability but does not provide mixnet-grade timing protection.